Model Performance Report

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System Overview

A snapshot of the current Piezo.AI platform state, including datasets loaded, models trained, predictions made, and database utilization.

Metric	Value
Datasets	3
Total Material Rows	287
Trained Models	6
Predictions Made	9
Training Jobs	8
Database Size	8.3 MB

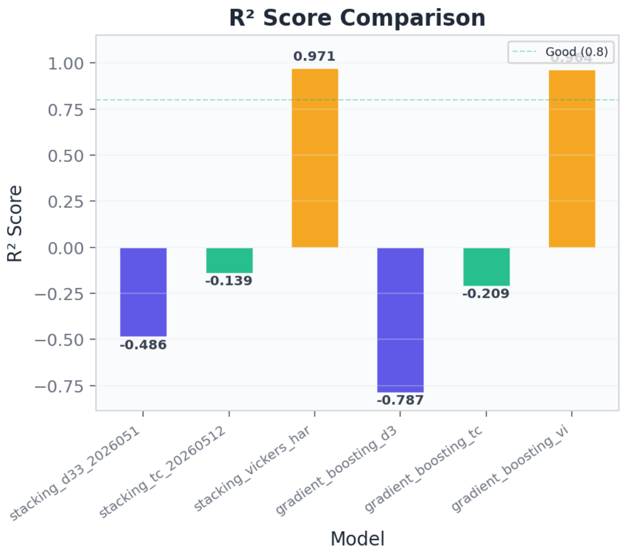
Trained Models Summary

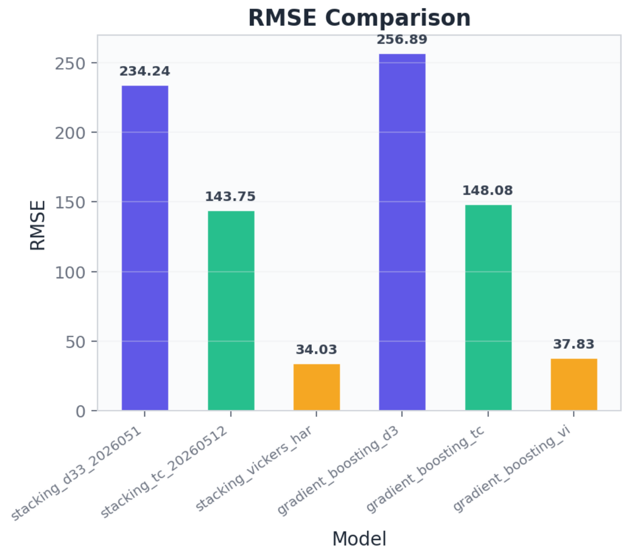
Overview of all trained ML models with their target property, algorithm, R^2 goodness-of-fit score, RMSE prediction error, and training/test split sizes.

Name	Target	Algorithm	R^2	RMSE	Train	Test
stacking_d33_20260512_093823	d33	stacking	**-0.4857**	234.24	24	6
stacking_tc_20260512_093823	Tc	stacking	**-0.1388**	143.75	24	6
stacking_vickers_hardness_20	HV	stacking	**0.9712**	34.03	24	6
gradient_boosting_d33_202605	d33	gradient_boosting	**-0.7868**	256.89	24	6
gradient_boosting_tc_2026051	Tc	gradient_boosting	**-0.2085**	148.08	24	6
gradient_boosting_vickers_ha	HV	gradient_boosting	**0.9644**	37.83	24	6

R^2 and RMSE Comparison

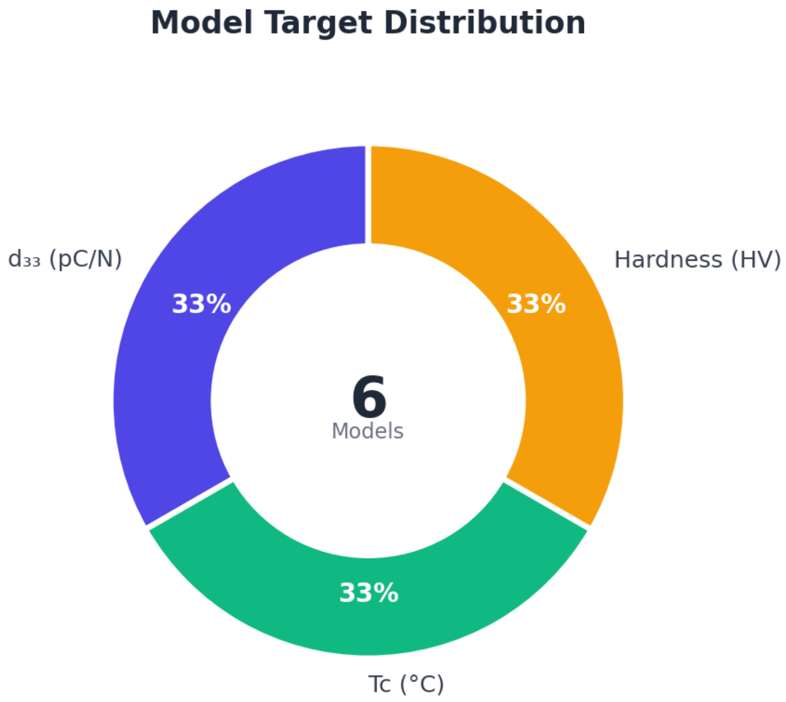
Side-by-side comparison of R^2 (coefficient of determination) and RMSE (root mean square error) across all trained models. Higher R^2 and lower RMSE indicate better model accuracy and reliability for material property predictions.





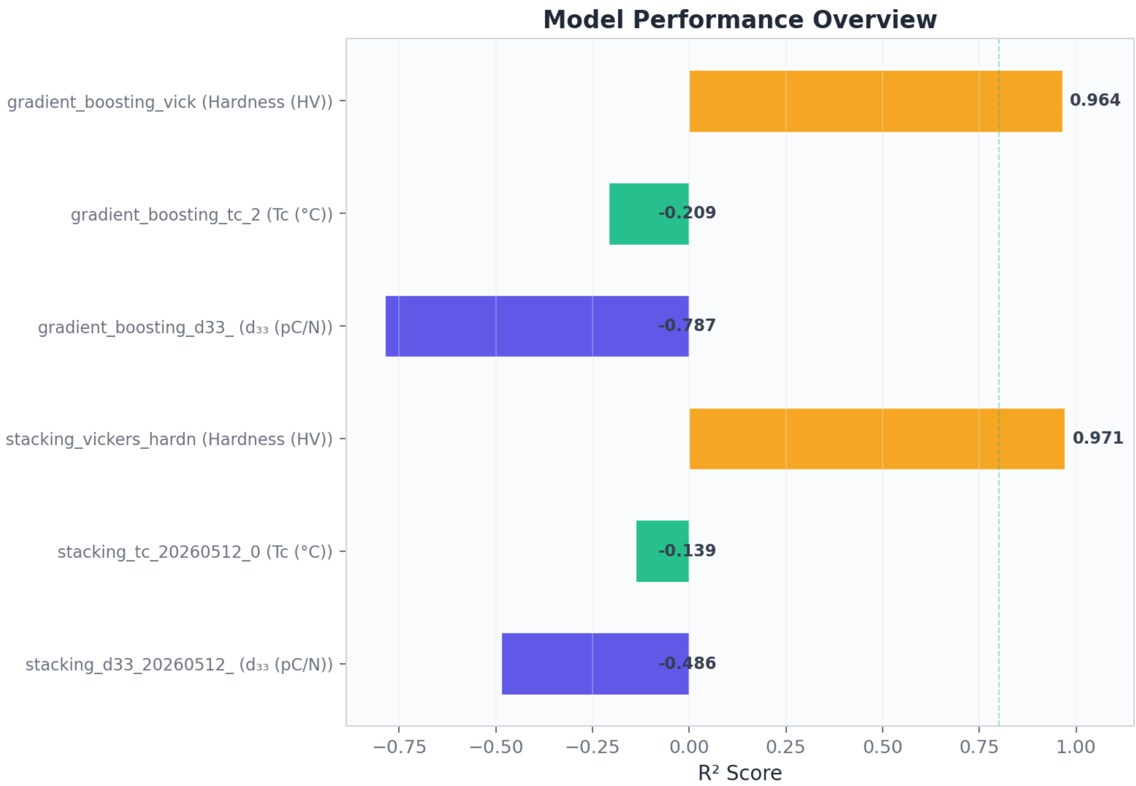
Model Target Distribution

Distribution of trained models across target properties (d■■, Tc, Hardness). Shows the proportion of models dedicated to predicting each material property, indicating research focus and coverage balance.



Performance Overview

Visual comparison of model performance via R^2 scores. The dashed line at 0.8 represents the threshold for reliable predictive quality. Models exceeding this threshold are considered suitable for guiding experimental material synthesis.



AI Performance Analysis

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The current training results for the Piezo.AI platform demonstrate a stark contrast in predictive performance between mechanical and functional properties. The Vickers hardness models achieved excellent results with R2 values exceeding 0.96, indicating that the current feature set effectively captures the relationship between material composition and mechanical durability. In contrast, the models for the piezoelectric charge coefficient and Curie temperature are performing poorly, with negative R2 values suggesting they are less accurate than a simple mean baseline. This failure to predict d33 and Tc is a significant hurdle for material discovery, as these properties are the primary drivers for piezoelectric application viability and thermal stability.

The disparity in performance is likely rooted in the complexity of the underlying physics and the current limitations of the training data. While mechanical hardness often follows relatively linear trends based on atomic packing and bond strength, piezoelectricity and phase transitions are highly sensitive to subtle structural distortions and electronic configurations. The high root mean square error values for d33 and Tc indicate that the models cannot currently distinguish between high-performance ceramics and low-response materials. For a discovery platform to be effective, these functional property models must reach an R2 of at least 0.80 to provide meaningful guidance for experimental synthesis.

A critical observation regarding data quality is the extremely small sample size used for model training. Although the system contains 287 material rows, the models were trained on only 24 samples with a 6-sample test set. This represents a significant underutilization of available data and is likely the primary cause of the poor generalization observed in the d33 and Tc models. Machine learning algorithms, particularly ensemble methods like gradient boosting and stacking, require substantially more data to map the non-linear multidimensional space of piezoelectric ceramics. The current models are likely suffering from high variance and are unable to learn the complex interactions required for accurate predictions.

To improve model accuracy, it is recommended to first integrate the full dataset of 287 rows into the training process using a k-fold cross-validation approach to ensure statistical robustness. Additionally, the feature engineering strategy should be expanded to include domain-specific descriptors such as the Goldschmidt tolerance factor, octahedral tilting angles, and electronegativity differences, which are known to influence the Curie temperature and piezoelectric response. Finally, a thorough data cleaning phase is necessary to ensure that the d33 and Tc values are consistent across the three datasets, as discrepancies in measurement techniques or units can introduce noise that prevents the models from converging on a valid solution.

AI Material Insight

The bulk KNaNbO3 ceramic is well suited for high temperature structural sensors and energy harvesting devices due to its high Curie temperature and significant hardness. Its piezoelectric coefficient of 149.3 pC/N places it firmly in the transducer range while maintaining structural integrity for industrial environments. In contrast, the KNaNbO3 composite variants, particularly the ZnSbO0.5 modified version, are ideal for high performance actuators and sensitive transducers. The KNaNbO3-ZnSbO0.5 composite exhibits a remarkably high d33 of 320.7 pC/N, making it a prime candidate for precision positioning systems and medical ultrasound applications where high displacement is required. The low hardness values of the composites suggest they may be integrated into flexible electronics or damping systems where mechanical compliance is beneficial.

These materials represent a significant advancement over traditional lead free alternatives like BaTiO3, which typically suffers from a low Curie temperature near 120 degrees Celsius. The predicted Curie temperatures exceeding 420 degrees Celsius for all samples allow these KNN based materials to compete with PZT in high temperature environments where thermal stability is critical. The transition from bulk KNaNbO3 to the ZnSbO0.5 modified composite demonstrates a clear trade off between mechanical hardness and piezoelectric performance. While the bulk ceramic offers superior structural robustness with a hardness of 456.6, the addition of ZnSbO0.5 significantly enhances the piezoelectric response by more than doubling the d33 value. This enhancement likely stems from the formation of a multiphase boundary or lattice distortion that facilitates domain switching, although it comes at the cost of reduced mechanical hardness. This relationship highlights the effectiveness of chemical modification and composite engineering in tailoring KNN based ceramics to meet the demands of high performance lead free electronics.

Prediction Insights

Predicted material properties for selected formulas, including d■■ piezoelectric coefficient, Curie temperature (Tc), Vickers hardness, and the top suggested use case based on the combination of predicted properties.

Formula	d_{33}	Tc	HV	Status
KNaNbO3	149.3	435.1	456.6	success
KNaNbO3-ZnSbO0.5	320.7	422.8	56.1	success